

Compute the PageRank scores on the given dataset

- Dataset: Data.txt
- The format of the lines in the file is as follow:

FromNodeID ToNodeID

- In this project, you need to report the **Top 100 NodeID** with their PageRank scores. You can choose different parameters, such as the teleport parameter, to compare different results.
- One result you must report is that when setting the teleport parameter to 0.85.
- In addition to the basic PageRank algorithm, you need to **optimize your memory use** via Block Matrix, Sparse Matrix and some other approaches.
- Deadline: 2026.5.15

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Specific Requirement:

- Language: You can use C/C++/Python.
- For C/C++, It's recommended to code in a single file and minimize the use of third-party libraries. You'd better to use gcc or g++ for compiling your code and give the compilation parameters in *compile-parameter.txt* with the following format.

```
g++ -fopenmp PageRank.cpp -o PageRank.exe
```
- For Python, It's recommended to use commonly-used libraries or package (e.g. numpy and scipy is accepted). To confirm the availability of a package, you can email webbigdata@163.com . It's recommended to code in a single file. Ensure the entry of your code should be named as 'main.py'.

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Specific Requirement:

- Consider of dead-ends and spider-traps
- Optimize your memory use **as much as possible**.
- You can optimize the memory use in other ways, but **Block Matrix and Sparse Matrix**(<https://zhuanlan.zhihu.com/p/557231877>) **optimization is compulsorily required**
- Your program is required to iterate until reaching convergence
- Using existing PageRank API is prohibited. (e.g. *networkx.pagerank* in Python)
- Your program need to read *Data.txt*, conduct PageRank and Return the Top-100 nodes in *Res.txt* with its score as following format:

NodeID Score

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Specific Requirement:

- You'd better make sure the maximum memory use during the whole life of your program should be lower than **80MB**.
- You can't to sacrifice too much time performance to minimize your memory usage. The program need to complete its runtime under **60s**.
- Maybe you can use some Parallel or/and Distributed Computing techniques. **But it's not the main content of this assignment.**

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Submitting:

- webbigdata@163.com
- *Report*: Include but not limited: description of dataset, key code details, **how to optimize the memory usage** and result with analysis
- *Code*: Meet the above requirements, Input Dataset is not required
- *Result*: Named as *Res.txt* and Meet the above requirements with a teleport parameter of 0.85
- *Executable File*: Compile with all **Static link library**.
 - For C/C++, maybe you can use `[-static]` setting in gcc/g++
 - For Python, using third-party packages to generate and integrate the package used in your code.
 - **Ensure your code can run in other computers (and mainstream OS).** (You can ask LLMs or TAs for more).

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Submitting:

- Indicate team division of work and contributions.
- If no team division indicated, whole group will be given the same score.
- Compress your submission (zip format), with following naming:
2000000_小张_2111111_小王_2222222_小林_第一次作业.zip
- Deadline: 12:00(UTC+8), 2026.5.15
- 10% of the points will be deducted for each day of late submission.
- **Submitting just one or two days before DDL is not recommended.**

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Hints:

- You **WILL NOT** get higher score only because you complete it solely (we recommend to complete it as a teamwork).
- You **CAN** use LLM for this work, **but purely AI-gen Report is explicitly prohibited.**
- **Failure to comply with the format will result in failure to grade.**
- **Code plagiarism and academic fraud are absolutely intolerable.**

Supplementary Materials

南开大学教务部文件 教字〔2025〕5号 南开大学本科教学人工智能工具使用规范（试行）

第八条 允许使用的范围。在征得任课教师或指导教师同意的前提下，且 AI 工具辅助完成的内容不会对课程作业、毕业论文（设计）、实验报告等的核心内容和创新能力考查造成影响时，学生可以在以下范围内合理使用 AI 工具。

1. 允许使用 AI 工具进行文献检索、资料收集、关键词推荐和文献管理，但须确保所引用文献的真实性和可靠性。

2. 允许使用 AI 工具进行语法检查、拼写校正、格式排版等基础性、非创造性工作，但需标注生成内容并限制比例。语言类学科专业应对使用标准另行明确限定。

3. 允许使用 AI 工具进行基于真实数据的图表制作、辅助制图，但应确保最终图表的原创性。

4. 允许使用 AI 工具进行程序代码的辅助编写、调试和纠错，所有代码或算法须经过作者审核和测试。

第九条 禁止使用的范围。

1. 禁止使用 AI 工具进行任何涉及敏感信息（如个人隐私、商业秘密等）或涉及保密的内容创作，禁止将涉密内容上传至任何 AI 平台。

2. 禁止使用 AI 工具生成或篡改原始核心论点、实验数据、调研数据，禁止使用 AI 工具生成基于虚假研究数据的图表、图片。

3. 禁止使用 AI 工具进行代写、抄袭、剽窃等学术不端行为。

4. 禁止在以上机考试、结课论文、笔试等形式开展的期中、期末测试中使用 AI 工具。特殊情况确需使用的，任课教师应提前予以说明。

5. 禁止使用 AI 工具直接替代任课教师或指导教师进行指导、评审、评价等教学工作，以确保学习效果评价的客观性和学术严谨性。